CODEFORCES
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PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

C. Fast Power Drift

time limit per test: 1 second
memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: pow, open, file, creat(, fstream, thread, process, system(, exec(

You are given two integers ***a*** and ***b***. Calculate $a^b \ mod \ 107$.

Input

The input file contains two integers ***a*** $(1 \leq a \leq 10^4)$ and ***b*** $(1 \leq b \leq 10^{12})$.

Output

Print one integer — the result of $a^b \ mod \ 107$.

Examples

input	Copy
100 3	

output	Copy
85	

input	Copy
100 5	

output	Copy
99	

input	Copy
10000 1000000000000	

output	Copy
27	

CSE221 Assignment 03 Fall 2026

Contest is running

6 days

Contestant

→ **Submit?**

Language: Java 21 64bit

Choose file: Choose File No file chosen

Verify you are human

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